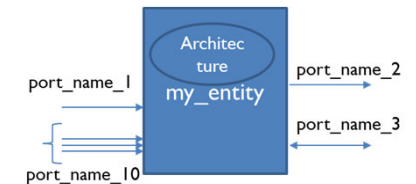
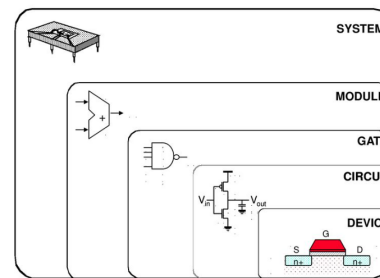
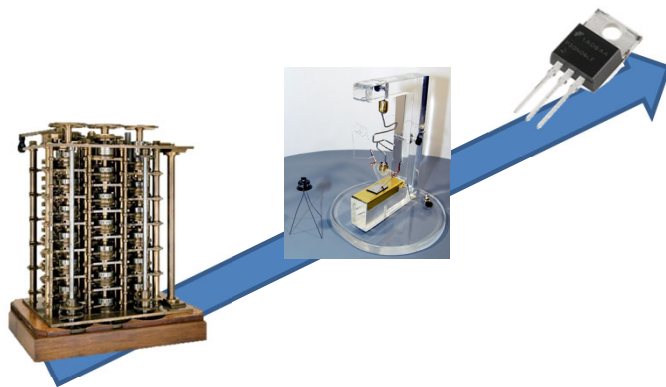


Lecture 12:

Q&A and course evaluation

By: Emad Samuel Malki Ebeid

Summary of Lecture I

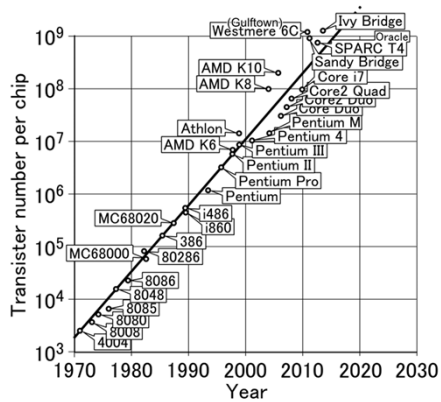


```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

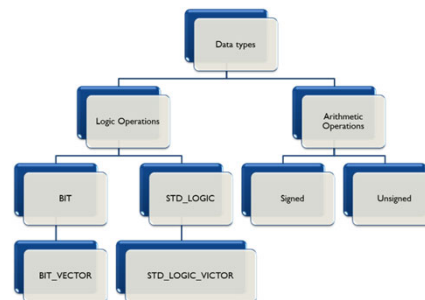
```
entity AND_Gate is
  Port ( A : in STD_LOGIC;
        B : in STD_LOGIC;
        C : out STD_LOGIC);
end AND_Gate;
```

```
architecture Behavioral of AND_Gate is
```

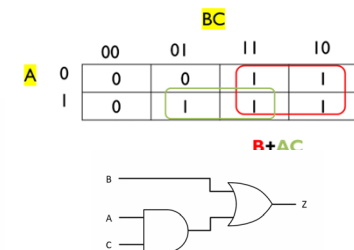
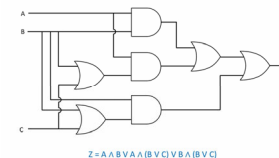
```
begin
  C <= A and B;
end Behavioral;
```



VHDL



Character	Value
'U'	uninitialized
'X'	strong drive, unknown logic value
'0'	strong drive, logic zero
'1'	strong drive, logic one
'Z'	high impedance
'W'	weak drive, unknown logic value
'L'	weak drive, logic zero
'H'	weak drive, logic one
'-'	don't care



Summary of lecture 2

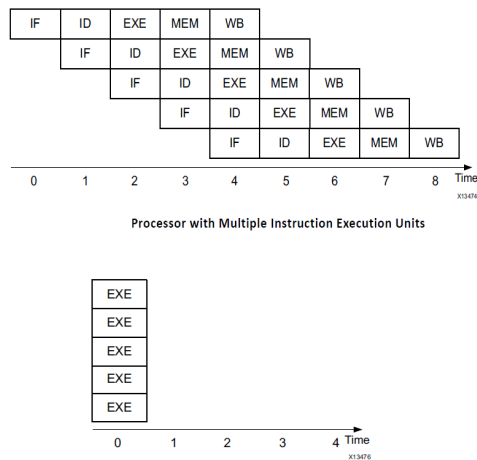
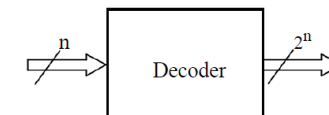
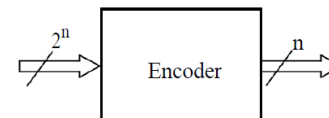
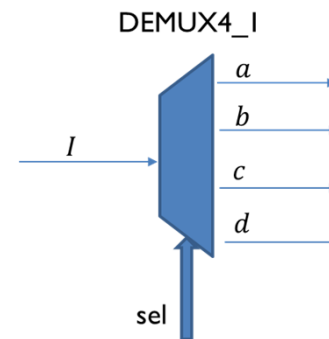


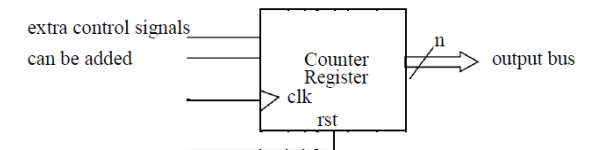
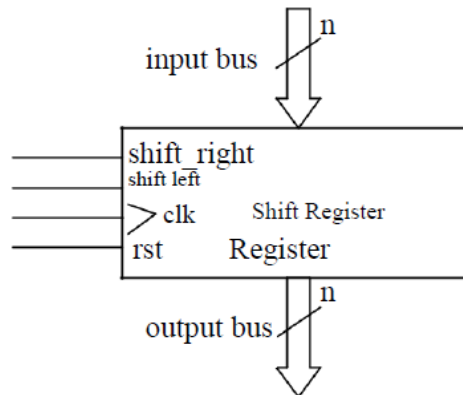
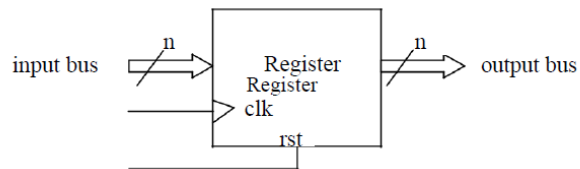
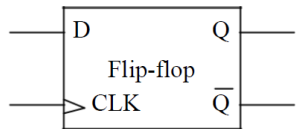
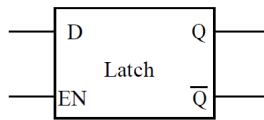
Figure 3-4: FPGA with Multiple Instruction Execution Units

- Concurrent and Sequential statements
 - Conditional signal assignment (when)
 - Selected Signal Assignment (with select)
 - Process statement (if/else)
 - Process statement (case)



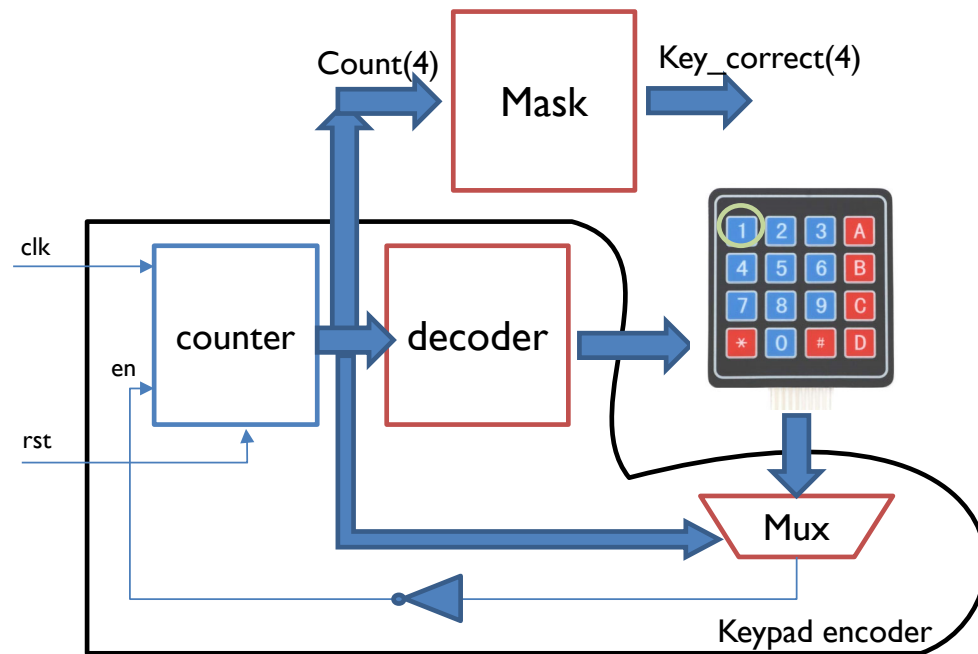
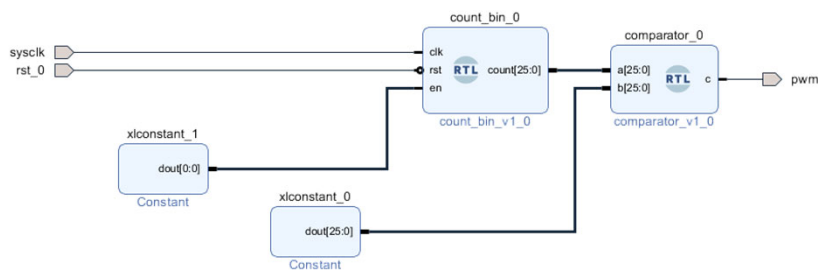
Summary of lecture 3

Signal and Variables

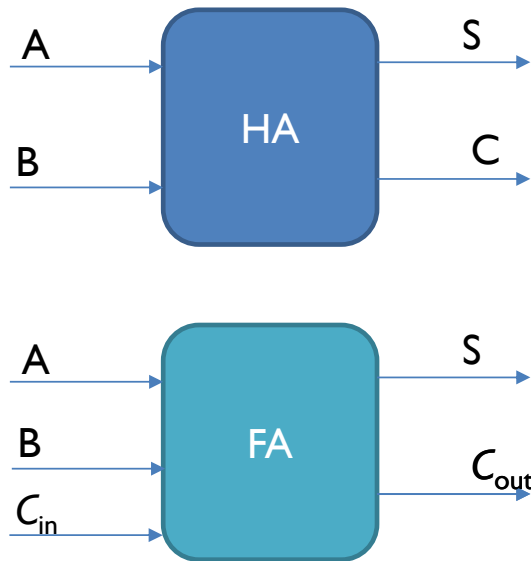


Summary of lecture 4

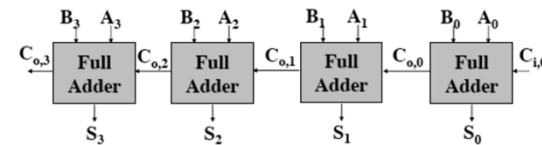
Operators	VHDL
Equality & Relational	=
	/=
	<
	<=
	>
	>=
Logical	not
	and
	or



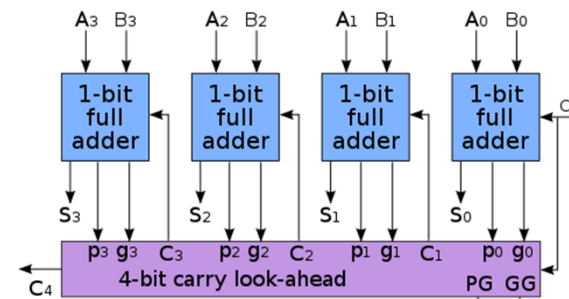
Summary of lecture 5



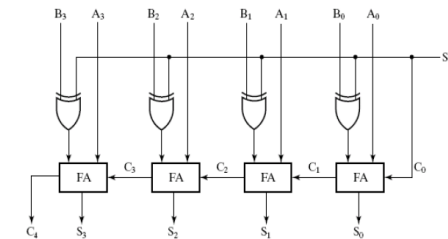
■ Ripple-carry adder



■ Carry-look-ahead adder

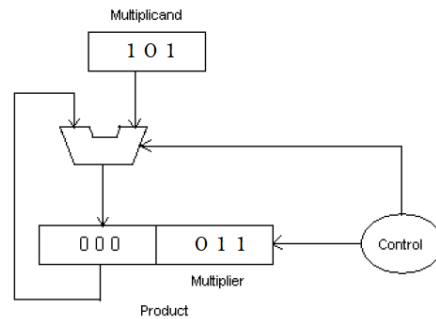


■ Full Adder/Subtractors

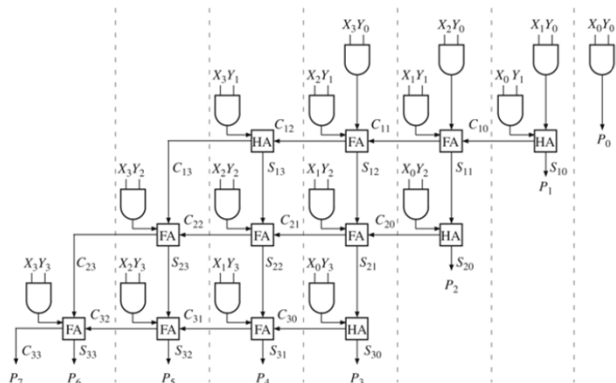


Summary of lecture 5

Multiplier

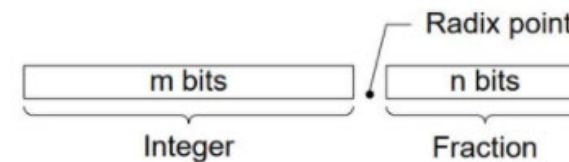


Array Multiplier

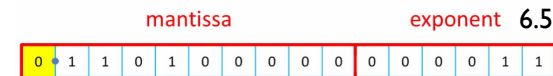


Dividers

- Clk divider
- Division by subtraction
- Division by multiplication
- Fixed point

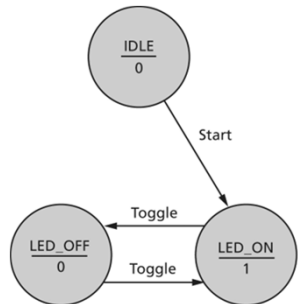


Floating point

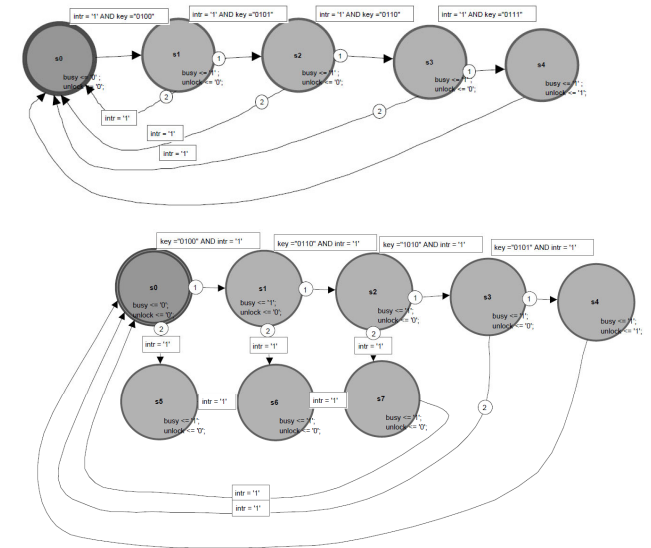
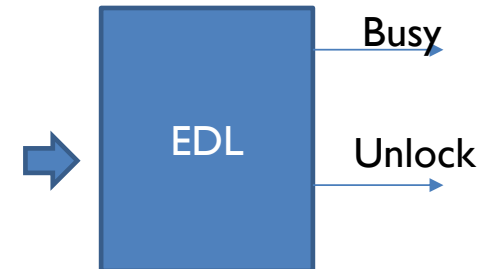
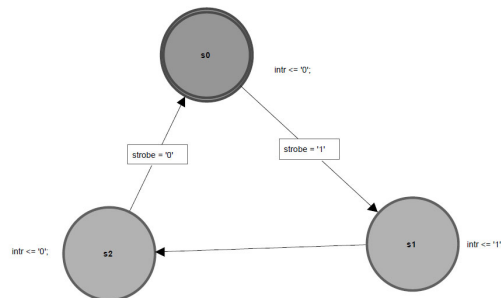
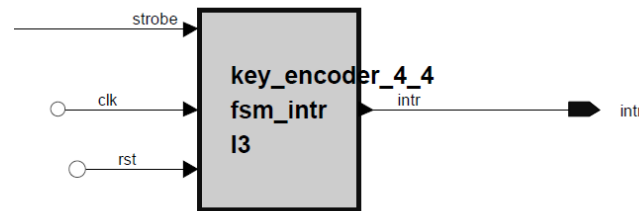
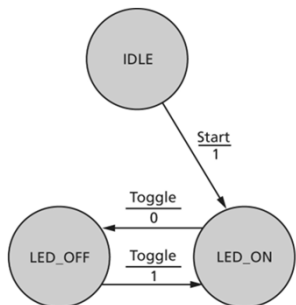


Summary of lecture 6

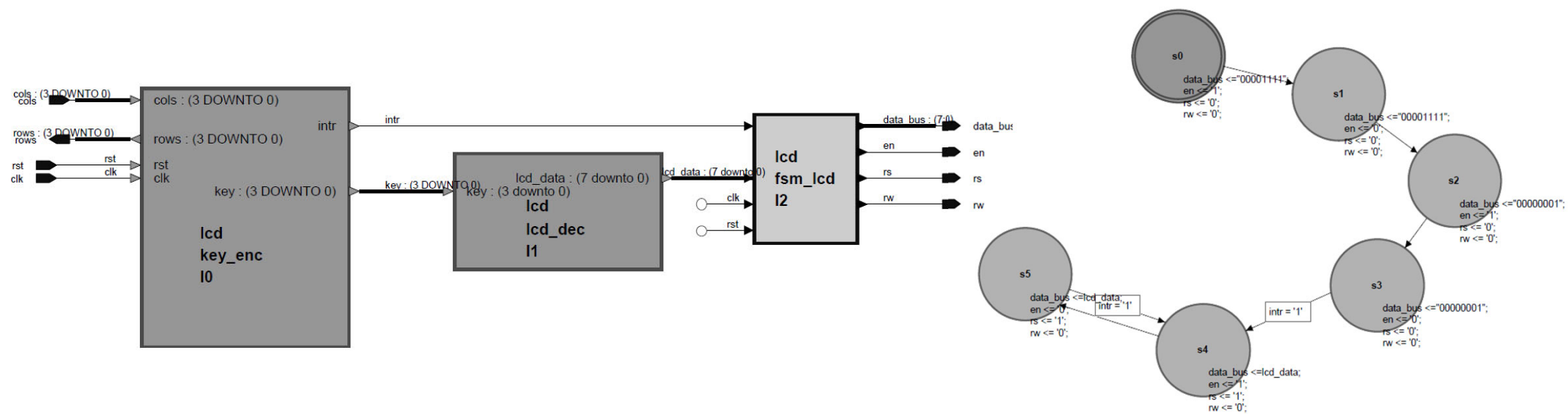
Moore



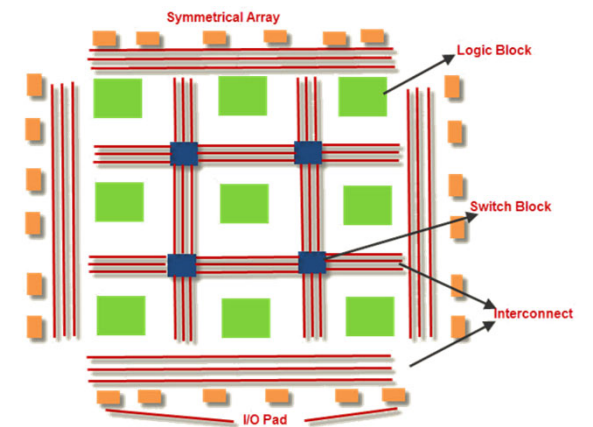
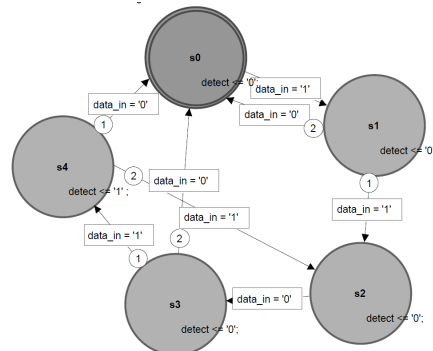
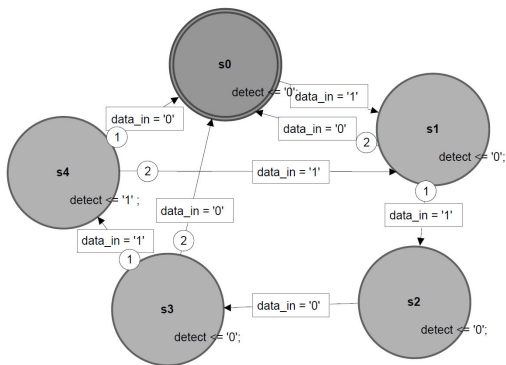
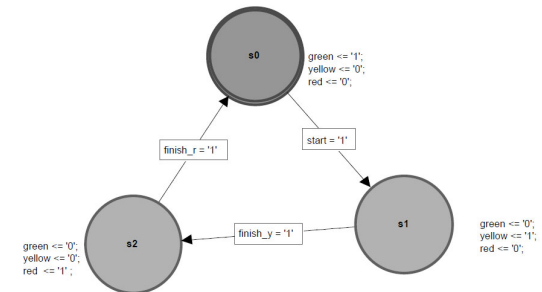
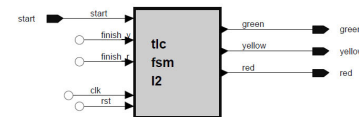
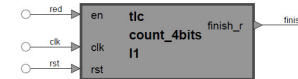
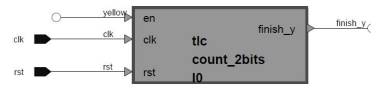
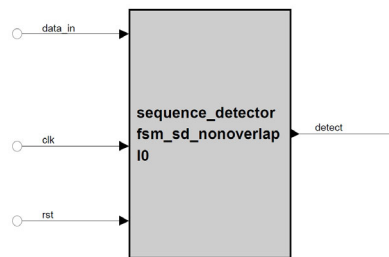
Mealy



Summary of lecture 6

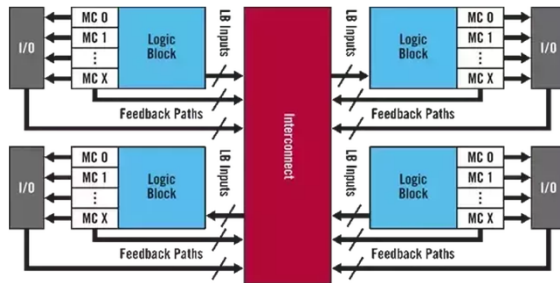


Summary of lecture 7



Summary of Lecture 8

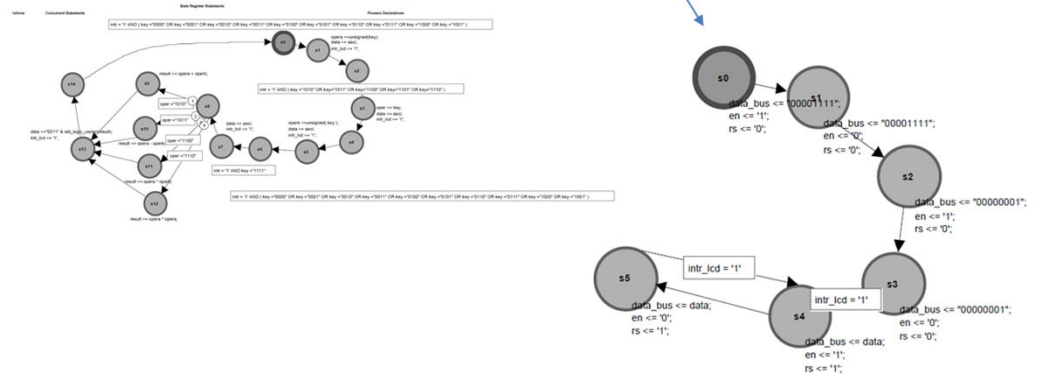
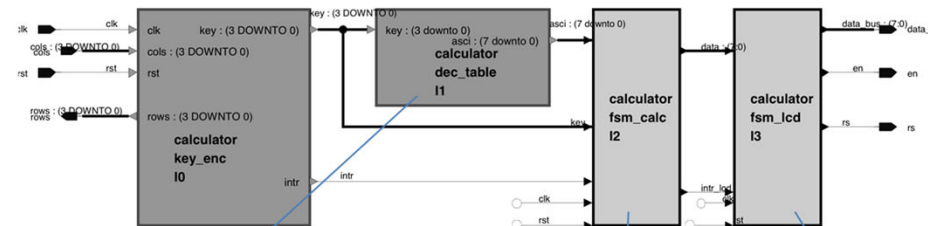
■ CPLD



```

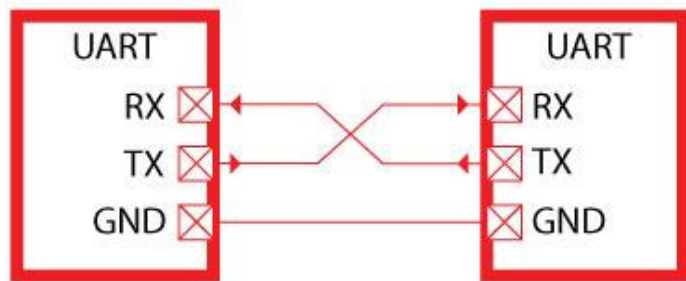
ENTITY dec_table IS
-- Declarations
  port(key: in std_logic_vector(3 downto 0);
        asci : out std_logic_vector(7 downto 0));
END dec_table ;

-- hds interface_end
ARCHITECTURE rtl OF dec_table IS
BEGIN
  process (key)
  begin
    if (key="0000") then asci<="00110000";
    elsif (key="0001") then asci<="00110001";--0
    elsif (key="0010") then asci<="00110010";
    elsif (key="0011") then asci<="00110011";
    elsif (key="0100") then asci<="00110100";
    elsif (key="0101") then asci<="00110101";
    elsif (key="0110") then asci<="00110110";
    elsif (key="0111") then asci<="00110111";
    elsif (key="1000") then asci<="00111000";
    elsif (key="1001") then asci<="00111001";--9
    elsif (key="1010") then asci<="00110101";--
    elsif (key="1011") then asci<="00101011";--
    elsif (key="1100") then asci<="00101010";--
    elsif (key="1101") then asci<="00101111";--2
    elsif (key="1110") then asci<="00101110";--
    else asci<="00111101"; --
    end if;
  end process;
END rtl;
  
```

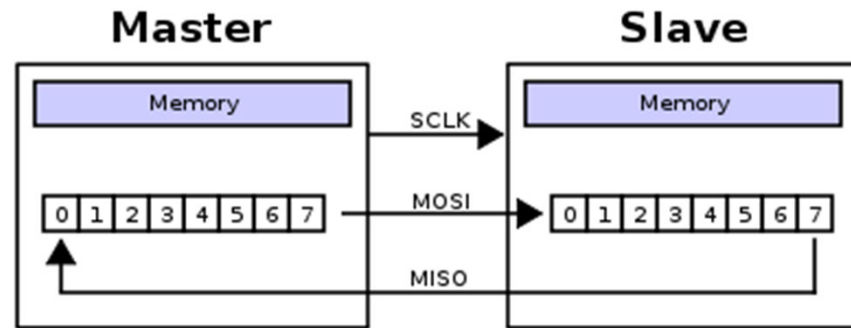


Summary of lecture 9

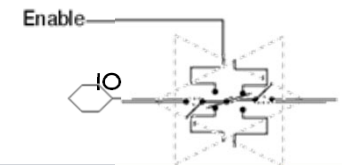
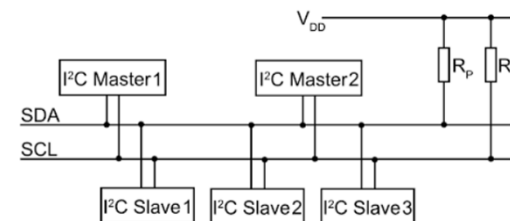
UART



SPI



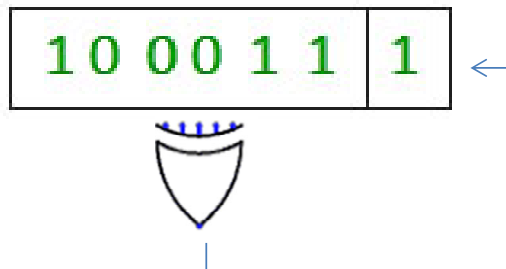
I2C



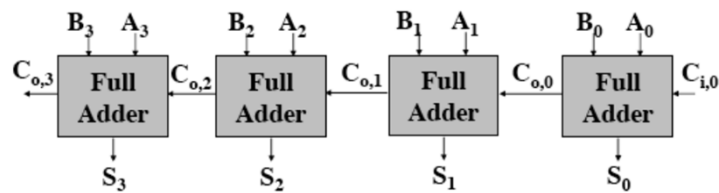
Summary of lecture 9

Error detection:

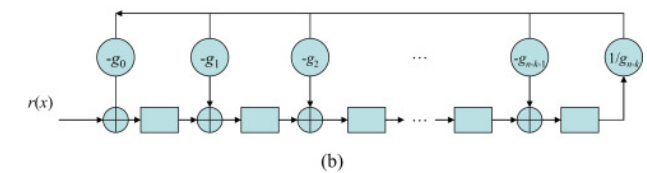
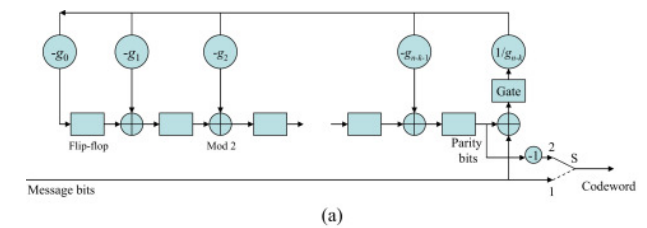
- Parity check



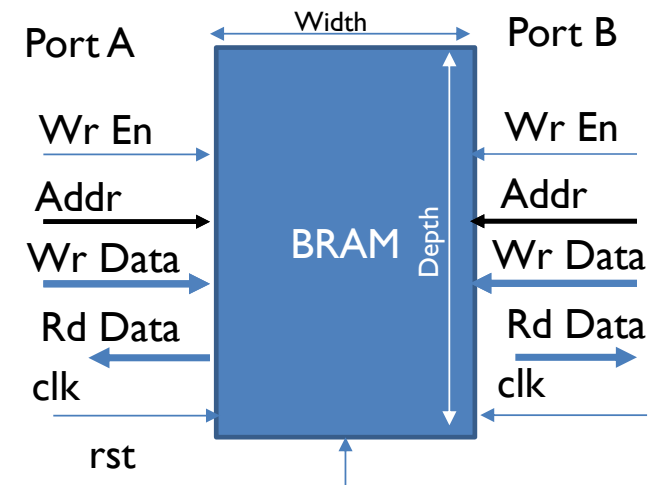
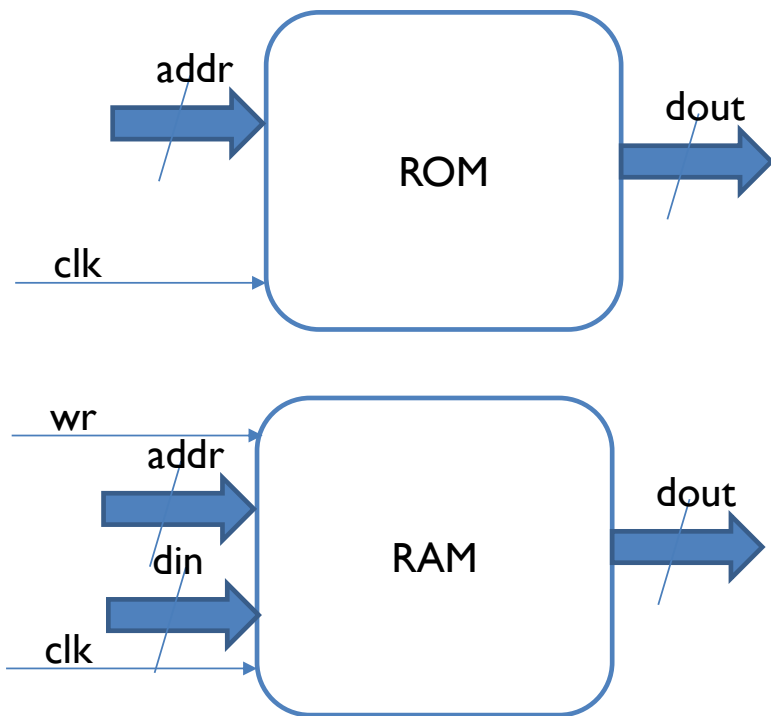
Checksum



CRC

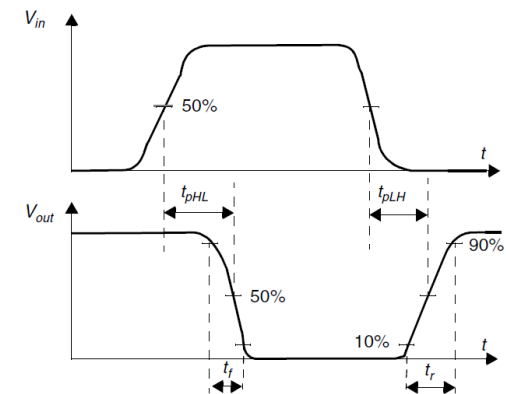
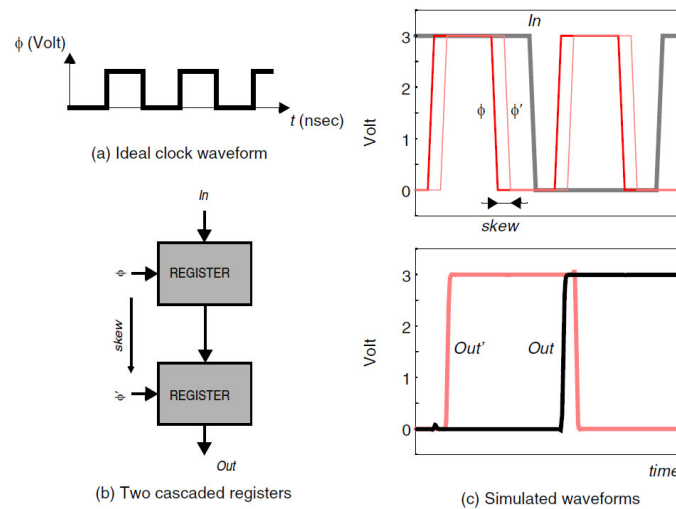


Summary of lecture 10



Summary of lecture I I

- Quality Metrics of a Digital Design
- Performance
- Propagation delay
- Metastable
- Clock Skew



Course Feedback

SDU Evaluation

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Exam setup

- Prepare a few slides to present your 3 assignments.
- During the exam, you will draw one of 3 cards that each represent an assignment to present for 5 minutes
- Then Q&As will take place for 10 minutes, see the next slides for inspiration.

Samples of the exam questions

- Define concurrent and sequential statements
- Which are the differences between conditional signal assignment and selected signal assignment
- What is the difference between a D-Latch and a D-Flip / Flop?
- Difference between signals and variables.
- Explain how a counter/shift register works?
- How do keypad encoders work?
- Explain how half/full adders, subtraction, multipliers, and dividers work?
- What's the difference between a Mealy and a Moore Machine?
- How EDL works

- Explain the FPGA design flow (functional and timing, simulation, synthesis, place, and route, etc.).
- What's the difference between CPLDs and FPGAs?
- Explain RAM/ROM and BRAM?
- How do UART, SPI, and I2C work, and what are the main differences between them?
- Questions related to the assignments



Master's course: Embedded Systems Course

<https://odin.sdu.dk/sitecore/index.php?a=searchfagbesk&bbcourseid=T550059101-I-E21>

CPU/FPGA, HLS, AI

Good Luck with the exam!

